

# Zishen Wan

## Assistant Professor of Computer Science

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### ACADEMIC APPOINTMENTS

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- 2026-present **Columbia University, New York, NY, USA**  
Assistant Professor of Computer Science, Department of Computer Science
- 2025-2026 **Harvard University, Cambridge, MA, USA**  
Postdoctoral Fellow, John A. Paulson School of Engineering and Applied Sciences (SEAS)  
Advisor: Prof. Vijay Janapa Reddi

### EDUCATION

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- 2020-2025 **Georgia Institute of Technology, Atlanta, GA, USA**  
Ph.D., School of Electrical and Computer Engineering (ECE)  
*Dissertation: Tailored Computing: Cross-Layer System, Architecture, and Silicon Co-Design for Physical Intelligence*  
Advisors: Prof. Arijit Raychowdhury and Prof. Tushar Krishna
- 2018-2020 **Harvard University, Cambridge, MA, USA**  
M.S., John A. Paulson School of Engineering and Applied Sciences (SEAS)
- 2014-2018 **Harbin Institute of Technology, Harbin, China**  
B.E. (Honors), Electrical Engineering

### RESEARCH AREAS

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Computer Architecture; VLSI and Systems-on-Chip; Machine Learning Systems; Hardware-Software Co-Design; Neuro-Symbolic AI; Embodied and Physical AI; Reliable AI Systems; and Agentic AI for Computing System Design.

My research spans two complementary directions: (1) **computing system design for AI**, developing efficient and reliable systems, architectures, and silicon for neuro-symbolic, embodied, and physical intelligence; and (2) **AI for computing system design**, using agentic AI to co-design and optimize software, architecture, and hardware.

### HONORS AND AWARDS

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- 2026 **ACM SIGDA Outstanding Ph.D. Dissertation Award**
- 2026 **WAIC Yunfan Rising Star Award** (15 awardees worldwide, under age 30)
- 2026 **ACM FCCM Outstanding Ph.D. Dissertation Award**
- 2026 **Colonel Oscar P. Cleaver Ph.D. Dissertation Award**, Georgia Tech
- 2025 **1st Place**, DAC Ph.D. Forum
- 2025 **Best Paper Award**, DARPA/SRC JUMP 2.0 Program
- 2025 **Best Poster Award**, DARPA/SRC JUMP 2.0 CoCoSys Center
- 2025 **Baidu Ph.D. Fellowship** (10 awardees worldwide)
- 2024 **Cyber-Physical Systems Rising Star**
- 2024 **3rd Place**, ACM/SIGMICRO Student Research Competition at MICRO
- 2024 **Best Paper Award**, DARPA/SRC JUMP 2.0 Program
- 2024 **Best Presentation Award**, Semiconductor Research Corporation TECHCON
- 2024 **Best Poster Award**, DARPA/SRC JUMP 2.0 CoCoSys Center
- 2023 **Machine Learning and Systems Rising Star**
- 2023 **IEEE Micro Top Picks**
- 2023 **Best Paper Award**, IROS Robotics Benchmarking Workshop
- 2023 **Best Poster Award**, IBM IEEE AI Compute Symposium
- 2023 **Roger P. Webb Graduate Research Assistant Excellence Award**, Georgia Tech
- 2022 **1st Place**, ACM/SIGBED Student Research Competition at ESWEEK
- 2022 **3rd Place**, ACM/SIGDA Student Research Competition at ICCAD (declined)
- 2022 **Qualcomm Fellowship**
- 2022 **CRNCH Ph.D. Fellowship**
- 2022 **DAC Young Fellow**, ACM/IEEE Design Automation Conference (DAC)
- 2021 **DAC Young Fellow and Best Presentation Award**, ACM/IEEE Design Automation Conference (DAC)

- 2021 **ACM SIGDA Research Highlights Nominee**
- 2020 **Best Paper Award**, ACM/IEEE Design Automation Conference (DAC)
- 2020 **Best Paper Award**, IEEE Computer Architecture Letters (CAL)
- 2018 **Chiang Chen Overseas Graduate Fellowship** (10 awardees worldwide)
- 2018 **Chunhui Innovation Achievement Award, First Class**, Harbin Institute of Technology

## BOOKS AND EDUCATIONAL RESOURCES

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- Machine Learning Systems**    **“Machine Learning Systems: Principles and Practices of Engineering AI Systems”**  
Vijay Janapa Reddi, Matthew Stewart, Ikechukwu Uchendu, Itai Shapira, Marcelo Rovai, Jayson Lin, Jeffrey Ma, Korneel Van den Berghe, **Zishen Wan**, Srivatsan Krishnan, Shvetank Prakash, Mark Mazumder, Colby Banbury, Jason Yik, Jessica Quaye, et al.  
*Open-source online book, pp. 1-2602, 2025; 27K+ GitHub stars*
- Synthesis Lectures on Computer Architecture**    **“Robotic Computing on FPGAs”**  
Shaoshan Liu, **Zishen Wan**, Bo Yu, and Yu Wang  
*Synthesis Lectures on Computer Architecture, Morgan & Claypool Publishers, pp. 1-218, 2021*

## PUBLICATIONS

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*Publications are listed in reverse chronological order. \* denotes equal contribution; my name is bold and underlined.*

### Preprints and Workshop Papers

- P1. arXiv’26**    **“ArchEval: Measuring AI Agents as Computer Architects”**  
Chenyu Wang\*, **Zishen Wan**\*, Jeffrey Ma, Shvetank Prakash, Zhenting Qi, Haebin Do, Andy Cheng, Arya Tschand, Jiahe Shi, Yilun Du, and Vijay Janapa Reddi  
*arXiv:2607.03601, 2026; presented at the Architecture 2.0 Workshop, ISCA 2026*
- P2. arXiv’26**    **“Dyserve: A Workflow-Aware Serving Layer for Agentic Applications”**  
Jiayi Qian, **Zishen Wan**, Hanchen Yang, Chun Tao, Souvik Kundu, and Tushar Krishna  
*arXiv:2607.02942, 2026; presented at the SCALE Workshop, ICML 2026*
- P3. DL4C’26**    **“Hawkeye: Hardware-Aware GPU Kernel Optimization with Minimal Supervision”**  
Arya Tschand, Kesavan Ramakrishnan, Alexander Ingare, Simon Guo, Jeffrey Jian Ma, **Zishen Wan**, Simran Arora, Azalia Mirhoseini, and Vijay Janapa Reddi  
*ICML Workshop on Deep Learning for Code (DL4C), 2026*
- P4. arXiv’26**    **“AgentDSE: Reasoning-Augmented Architectural Design Space Exploration”**  
Chenyu Wang, Jiahe Caroline Shi, David Kong, Duane S. Boning, **Zishen Wan**, Yilun Du, and Vijay Janapa Reddi  
*arXiv:2606.21836, 2026; MLArchSys Workshop, ISCA 2026*
- P5. arXiv’26**    **“FLARE: One-Shot PE-Level Fault Localization in Systolic Arrays via Algebraic Test Vectors”**  
Logashree Venkatasubramanian, **Zishen Wan**, and Viveck Cadambe  
*arXiv:2605.08594, 2026*
- P6. arXiv’26**    **“GenAI for Systems: Recurring Challenges and Design Principles from Software to Silicon”**  
Arya Tschand\*, Chenyu Wang\*, **Zishen Wan**\*, Andrew Cheng, Ioana Cristescu, Kevin He, Howard Huang, Alexander Ingare, Akseli Kangaslahti, Sara Kangaslahti, Theo Lebryk, Hongjin Lin, Jeffrey Jian Ma, Alexandru Meterez, Clara Mohri, Depen Morwani, Sunny Qin, Roy Rinberg, Paula Rodriguez-Diaz, Alyssa Mia Taliotis, Pernille Undrum Fathi, Rosie Zhao, Todd Zhou, and Vijay Janapa Reddi  
*arXiv:2602.15241, 2026*
- P7. arXiv’25**    **“ANNIE: Be Careful of Your Robots”**  
Yiyang Huang, Zixuan Wang, **Zishen Wan**, Yapeng Tian, Haobo Xu, Yinhe Han, and Yiming Gan  
*arXiv:2509.03383, 2025*
- P8. arXiv’25**    **“ADDT: A Digital Twin Framework for Proactive Safety Validation in Autonomous Driving Systems”**  
Bo Yu\*, Chaoran Yuan\*, **Zishen Wan**\*, Jie Tang, Fadi Kurdahi, and Shaoshan Liu  
*arXiv:2504.09461, 2025*
- P9. MLSys Wksp.’24**    **“Towards Cognitive AI Systems: A Survey and Prospective on Neuro-Symbolic AI”**  
**Zishen Wan**, Che-Kai Liu, Hanchen Yang, Chaojian Li, Haoran You, Yonggan Fu, Cheng Wan, Tushar Krishna, Yingyan Lin, and Arijit Raychowdhury  
*MLSys Workshop on Systems for Next-Generation AI Paradigms, 2024*

- P10. ICLR Wksp.'23** “Scaling Compute Is Not All You Need for Adversarial Robustness”  
Edoardo Debenedetti, Zishen Wan, Maksym Andriushchenko, Vikash Sehwal, Kshitij Bhardwaj, and Bhavya Kailkhura  
*ICLR Workshop on Reliable and Responsible Foundation Models, 2023*
- P11. NVMW'22** “RRAM-ECC: Improving Reliability of RRAM-Based Compute In-Memory”  
Zishen Wan, Brian Crafton, Samuel Spetalnick, Jong-Hyeok Yoon, and Arijit Raychowdhury  
*Annual Non-Volatile Memories Workshop (NVMW), 2022*

#### Refereed Conference Publications

- C1. HPCA'26** “REASON: Accelerating Probabilistic Logical Reasoning for Scalable Neuro-Symbolic Intelligence”  
Zishen Wan, Che-Kai Liu, Jiayi Qian, Hanchen Yang, Arijit Raychowdhury, and Tushar Krishna  
*IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2026*
- C2. ASPLOS'26** “Compositional AI Beyond LLMs: System Implications of Neuro-Symbolic-Probabilistic Architectures”  
Zishen Wan, Hanchen Yang, Jiayi Qian, Ritik Raj, Joongun Park, Chenyu Wang, Arijit Raychowdhury, and Tushar Krishna  
*ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2026*
- C3. ASPLOS'26** “CREATE: Cross-Layer Resilience Characterization and Optimization for Efficient yet Reliable Embodied AI Systems”  
Tong Xie, Yijiahao Qi, Jinqi Wen, Zishen Wan, Yanchi Dong, Zihao Wang, Shaofei Cai, Yitao Liang, Tianyu Jia, Yuan Wang, Runsheng Wang, and Meng Li  
*ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2026*
- C4. ICLR'26** “SLM-Mux: Orchestrating Small Language Models for Reasoning”  
Chenyu Wang\*, Zishen Wan\*, Hao Kang, Emma Chen, Zhiqiang Xie, Tushar Krishna, Vijay Janapa Reddi, and Yilun Du  
*International Conference on Learning Representations (ICLR), 2026*
- C5. ICML'26** “QuArch: A Benchmark for Evaluating LLM Reasoning in Computer Architecture”  
Shvetank Prakash, Andrew Cheng, Arya Tschand, Mark Mazumder, Varun Gohil, Jeffrey Ma, Jason Yik, Zishen Wan, Jessica Quaye, Elisavet Lydia Alvanaki, Avinash Kumar, Chandrasis Mazumdar, Tuhin Khare, Alexander Ingare, Ikechukwu Uchendu, Radhika Ghosal, Abhishek Tyagi, Chenyu Wang, Andrea Mattia Garavagno, Sarah Gu, Alice Guo, Grace Hur, Luca Carloni, Tushar Krishna, Ankita Nayak, Amir Yazdanbakhsh, and Vijay Janapa Reddi  
*International Conference on Machine Learning (ICML), 2026*
- C6. MICRO'26** “Arborist: Algorithm-Hardware Co-Design for Fast and Efficient Motion Planning”  
Yaotian Liu, Lingyi Huang, Zishen Wan, Bo Yuan, Cheng Tan, and Jeff Zhang  
*IEEE/ACM International Symposium on Microarchitecture (MICRO), 2026*
- C7. DAC'26** “Efficient Mixture-of-Agents Serving via Tree-Structured Routing, Adaptive Pruning, and Dependency-Aware Prefill-Decode Overlap”  
Zijun Wang, Yijiahao Qi, Hanqiu Chen, Zishen Wan, Gongjin Sun, Dongyang Li, Shuyi Pei, and Cong (Callie) Hao  
*ACM/IEEE Design Automation Conference (DAC), 2026*
- C8. DATE'26** “FortiSky: Enhancing Adversarial and Bit-Error Robustness for Efficient and Secure Autonomous Systems”  
Zishen Wan, Karthik Swaminathan, Nandhini Chandramoorthy, Pin-Yu Chen, Tushar Krishna, Vijay Janapa Reddi, and Arijit Raychowdhury  
*Design, Automation and Test in Europe Conference (DATE), 2026*
- C9. DATE'26** “SATA: Sparsity-Aware Scheduling for Selective Token Attention”  
Zhenkun Fan, Zishen Wan, Che-Kai Liu, Ashwin Lele, Win-San (Vince) Khwa, Bo Zhang, Meng-Fan (Marvin) Chang, and Arijit Raychowdhury  
*Design, Automation and Test in Europe Conference (DATE), 2026*

- C10. FCCM'26** “Enabling Context-Switchable Monolithic 3D FPGA Design Using Bistable Ferroelectric Inverters”  
Faaiq Waqar, Matthew Chen, Zifan He, Zishen Wan, Minji Shon, Wei-Hsing Huang, Jason Cong, and Shimeng Yu  
*IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM)*, 2026
- C11. ISPASS'26** “Towards System-2 AI: Workloads and Characterizations of Energy-Based Models”  
Hanchen Yang, Jiayi Qian, Zishen Wan, Jingtian Dang, Ziwei Li, Yilun Du, and Tushar Krishna  
*IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, 2026
- C12. ICLAD'26** “LACE: Large Language Model Aided Multi-Agent Framework for Agile RISC-V Instruction Extension”  
Pingqing Zheng, Jiayin Qin, Fuqi Zhang, Zishen Wan, Shang Wu, Yu (Kevin) Cao, Caiwen Ding, and Yang (Katie) Zhao  
*IEEE International Conference on LLM-Aided Design (ICLAD)*, 2026
- C13. ASP-DAC'26** “HDLxGraph: Bridging Large Language Models and HDL Repositories via HDL Graph Databases”  
Pingqing Zheng, Jiayin Qin, Fuqi Zhang, Niraj Chitla, Zishen Wan, Shang Wu, Yu (Kevin) Cao, Caiwen Ding, and Yang (Katie) Zhao  
*Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026
- C14. HPCA'25** “CogSys: Efficient and Scalable Neurosymbolic Cognition System via Algorithm-Hardware Co-Design”  
Zishen Wan<sup>\*</sup>, Hanchen Yang<sup>\*</sup>, Ritik Raj<sup>\*</sup>, Che-Kai Liu, Ananda Samajdar, Arijit Raychowdhury, and Tushar Krishna  
*IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2025  
**Best Paper Award, DARPA/SRC JUMP 2.0, 2024; 3rd Place, ACM SIGMICRO Student Research Competition, 2024**
- C15. ASPLOS'25** “ReCA: Integrated Acceleration for Real-Time and Efficient Cooperative Embodied Autonomous Agents”  
Zishen Wan, Yuhang Du, Mohamed Ibrahim, Jiayi Qian, Jason Jabbour, Yang (Katie) Zhao, Tushar Krishna, Arijit Raychowdhury, and Vijay Janapa Reddi  
*ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2025  
**1st Place, DAC Ph.D. Forum, 2025; selected as an Industry-Academia Partnership Highlight**
- C16. ASPLOS'25** “OctoCache: Caching Voxels for Accelerating 3D Occupancy Mapping in Autonomous Systems”  
Peiqing Chen, Minghao Li, Zishen Wan, Yu-Shun Hsiao, Minlan Yu, Vijay Janapa Reddi, and Alan Liu  
*ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2025
- C17. MICRO'25** “RTGS: Real-Time 3D Gaussian Splatting SLAM via Multi-Level Redundancy Reduction”  
Leshu Li, Jiayin Qin, Jie Peng, Zishen Wan, Huaizhi Qu, Ye Han, Pingqing Zheng, Hongsen Zhang, Yu (Kevin) Cao, Tianlong Chen, and Yang (Katie) Zhao  
*IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2025
- C18. DAC'25** “NSFlow: An End-to-End FPGA Framework with Scalable Dataflow Architecture for Neuro-Symbolic AI”  
Hanchen Yang<sup>\*</sup>, Zishen Wan<sup>\*</sup>, Ritik Raj, Joongun Park, Ziwei Li, Ananda Samajdar, Arijit Raychowdhury, and Tushar Krishna  
*ACM/IEEE Design Automation Conference (DAC)*, 2025
- C19. DAC'25** “ReaLM: Reliable and Efficient Large Language Model Inference with Statistical Algorithm-Based Fault Tolerance”  
Tong Xie, Jiawang Zhao, Zishen Wan, Zuodong Zhang, Yuan Wang, Runsheng Wang, Ru Huang, and Meng Li  
*ACM/IEEE Design Automation Conference (DAC)*, 2025
- C20. ISPASS'25** “Generative AI in Embodied Systems: System-Level Analysis of Performance, Efficiency and Scalability”  
Zishen Wan, Jiayi Qian, Yuhang Du, Jason Jabbour, Yilun Du, Yang (Katie) Zhao, Arijit Raychowdhury, Tushar Krishna, and Vijay Janapa Reddi  
*IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, 2025

- C21. ISPASS'25** “SCALE-Sim v3: A Modular Cycle-Accurate Systolic Accelerator Simulator for End-to-End System Analysis”  
Ritik Raj, Sarbartha Banerjee\*, Nikhil Chandra Boopalam Srinivas\*, **Zishen Wan\***, Jianming Tong\*, Ananda Samajdar, and Tushar Krishna  
*IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, 2025
- C22. ICCAD'25** “HyDra: SOT-CAM Based Vector Symbolic Macro for Hyperdimensional Computing”  
Mizanur Nayan, Che-Kai Liu, **Zishen Wan**, Arijit Raychowdhury, and Azad J. Naeemi  
*IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2025
- C23. NeuS'25** “Efficient Processing of Neuro-Symbolic AI: A Tutorial and Cross-Layer Co-Design Case Study”  
**Zishen Wan**, Che-Kai Liu, Hanchen Yang, Ritik Raj, Arijit Raychowdhury, and Tushar Krishna  
*International Conference on Neuro-symbolic Systems (NeuS)*, 2025  
**Selected for oral presentation (top 3%)**
- C24. ISCAS'25** “Characterization and Mitigation of ADC Noise by Reference Tuning in RRAM-Based CIM”  
Ying-Hao Wei\*, **Zishen Wan\***, Brian Crafton, Samuel Spetalnick, and Arijit Raychowdhury  
*IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025
- C25. ASPLOS'24** “MulBERRY: Enabling Bit-Error Robustness for Energy-Efficient Multi-Agent Autonomous Systems”  
**Zishen Wan**, Nandhini Chandramoorthy, Karthik Swaminathan, Pin-Yu Chen, Kshitij Bhardwaj, Vijay Janapa Reddi, and Arijit Raychowdhury  
*ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2024  
**Best Poster Award, IBM IEEE AI Compute Symposium, 2023**
- C26. ASPLOS'24** “ORIANNA: An Accelerator Generation Framework for Optimization-Based Robotic Applications”  
Yuhui Hao, Yiming Gan, Bo Yu, Qiang Liu, Yinhe Han, **Zishen Wan**, and Shaoshan Liu  
*ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2024
- C27. DAC'24** “Algorithm-Hardware Co-Design of Distribution-Aware Logarithmic-Posit Encodings for Efficient DNN Inference”  
Akshat Ramachandran, **Zishen Wan**, Geonhwa Jeong, John Gustafson, and Tushar Krishna  
*ACM/IEEE Design Automation Conference (DAC)*, 2024
- C28. ICCAD'24** “Thinking and Moving: An Efficient Computing Approach for Integrated Task and Motion Planning in Cooperative Embodied AI Systems”  
**Zishen Wan**, Yuhang Du, Mohamed Ibrahim, Yang (Katie) Zhao, Tushar Krishna, and Arijit Raychowdhury  
*IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2024
- C29. ISPASS'24** “Towards Cognitive AI Systems: Workload and Characterization of Neuro-Symbolic AI”  
**Zishen Wan**, Che-Kai Liu, Hanchen Yang, Ritik Raj, Chaojian Li, Haoran You, Yonggan Fu, Cheng Wan, Ananda Samajdar, Yingyan Lin, Tushar Krishna, and Arijit Raychowdhury  
*IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, 2024  
**Best Poster Award, DARPA/SRC JUMP 2.0 CoCoSys Center, 2024**
- C30. DATE'24** “H3DFact: Heterogeneous 3D Integrated CIM for Factorization with Holographic Perceptual Representations”  
**Zishen Wan\***, Che-Kai Liu\*, Mohamed Ibrahim, Hanchen Yang, Samuel Spetalnick, Tushar Krishna, and Arijit Raychowdhury  
*Design, Automation and Test in Europe Conference (DATE)*, 2024  
**Best Presentation Award, SRC TECHCON, 2024**
- C31. ESWEEK'24** “Neuro-Symbolic Architecture Meets Large Language Models: A Memory-Centric Perspective”  
Mohamed Ibrahim, **Zishen Wan**, Haitong Li, Priyadarshini Panda, Tushar Krishna, Pentti Kanerva, Yiran Chen, and Arijit Raychowdhury  
*Embedded Systems Week (ESWEEK)*, 2024



- C32. ICRA'24** **“RobotPerf: An Open-Source, Vendor-Agnostic Benchmarking Suite for Evaluating Robotics Computing System Performance”**  
Victor Mayoral-Vilches, Jason Jabbour, Yu-Shun Hsiao, **Zishen Wan**, Alejandra Martinez-Farina, Martino Crespo-Alvarez, Matthew Stewart, Juan Manuel Reina-Munoz, Prateek Nagras, Gaurav Vikhe, Mohammad Bakhshalipour, Martin Pinzger, Stefan Rass, Smruti Panigrahi, Giulio Corradi, Niladri Roy, Phillip B. Gibbons, Sabrina M. Neuman, Brian Plancher, and Vijay Janapa Reddi  
*IEEE International Conference on Robotics and Automation (ICRA)*, 2024  
**Best Paper Award, IROS Robotics Benchmarking Workshop, 2023**
- C33. DAC'23** **“BERRY: Bit Error Robustness for Energy-Efficient Reinforcement Learning-Based Autonomous Systems”**  
**Zishen Wan**, Nandhini Chandramoorthy, Karthik Swaminathan, Pin-Yu Chen, Vijay Janapa Reddi, and Arijit Raychowdhury  
*ACM/IEEE Design Automation Conference (DAC)*, 2023
- C34. DATE'23** **“MAVFI: An End-to-End Fault Analysis Framework with Anomaly Detection and Recovery for Micro Aerial Vehicles”**  
Yu-Shun Hsiao\*, **Zishen Wan\***, Tianyu Jia, Radhika Ghosal, Abdulrahman Mahmoud, Arijit Raychowdhury, David Brooks, Gu-Yeon Wei, and Vijay Janapa Reddi  
*Design, Automation and Test in Europe Conference (DATE)*, 2023
- C35. ICCAD'23** **“SEE-MCAM: Scalable Multi-Bit FeFET Content Addressable Memories for Energy-Efficient Associative Search”**  
Shengxi Shou, Che-Kai Liu, Sanggeon Yun, **Zishen Wan**, Kai Ni, Mohsen Imani, X. Sharon Hu, Jianyi Yang, Cheng Zhuo, and Xunzhao Yin  
*IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023
- C36. DATE'23** **“Real-Time Fully Unsupervised Domain Adaptation for Lane Detection in Autonomous Driving”**  
Kshitij Bhardwaj, **Zishen Wan**, Arijit Raychowdhury, and Ryan Goldhahn  
*Design, Automation and Test in Europe Conference (DATE)*, 2023
- C37. ISSCC'23** **“A 73.53 TOPS/W 14.74 TOPS Heterogeneous RRAM In-Memory and SRAM Near-Memory SoC for Hybrid Frame and Event-Based Target Tracking”**  
Muya Chang, Ashwin Lele, Samuel Spetalnick, Brian Crafton, Shota Konna, **Zishen Wan**, Ashwin Bhat, Win-San Khwa, Yu-Der Chih, Meng-Fan Chang, and Arijit Raychowdhury  
*IEEE International Solid-State Circuits Conference (ISSCC)*, 2023
- C38. MICRO'22** **“Automatic Domain-Specific SoC Design for Autonomous Unmanned Aerial Vehicles”**  
Srivatsan Krishnan, **Zishen Wan**, Kshitij Bhardwaj, Paul Whatmough, Aleksandra Faust, Sabrina M. Neuman, Gu-Yeon Wei, David Brooks, and Vijay Janapa Reddi  
*IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2022  
**IEEE Micro Top Picks, Honorable Mention**
- C39. DAC'22** **“Improving Compute In-Memory ECC Reliability with Successive Correction”**  
Brian Crafton, **Zishen Wan**, Samuel Spetalnick, Jong-Hyeok Yoon, Wei Wu, Carlos Tokunaga, Vivek De, and Arijit Raychowdhury  
*ACM/IEEE Design Automation Conference (DAC)*, 2022
- C40. ISPASS'22** **“Roofline Model for UAVs: A Bottleneck Analysis Tool for Onboard Compute Characterization of Autonomous Unmanned Aerial Vehicles”**  
Srivatsan Krishnan, **Zishen Wan**, Kshitij Bhardwaj, Ninad Jadhav, Aleksandra Faust, and Vijay Janapa Reddi  
*IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, 2022
- C41. ICCAD'22** **“Analyzing and Improving Resilience and Robustness of Autonomous Systems”**  
**Zishen Wan**, Karthik Swaminathan, Nandhini Chandramoorthy, and Arijit Raychowdhury  
*IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2022
- C42. AICAS'22** **“Robotic Computing on FPGAs: Current Progress, Research Challenges, and Opportunities”**  
**Zishen Wan**, Ashwin Lele, Bo Yu, Shaoshan Liu, Yu Wang, Vijay Janapa Reddi, Cong (Callie) Hao, and Arijit Raychowdhury  
*IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, 2022

- C43. CICC'22** “An Energy-Efficient and Runtime-Reconfigurable FPGA-Based Accelerator for Robotic Localization Systems”  
Qiang Liu\*, **Zishen Wan\***, Bo Yu\*, Weizhuang Liu, Shaoshan Liu, and Arijit Raychowdhury  
*IEEE Custom Integrated Circuits Conference (CICC)*, 2022
- C44. DATE'22** “FRL-FI: Transient Fault Analysis for Federated Reinforcement Learning-Based Navigation Systems”  
**Zishen Wan**, Aqeel Anwar, Abdulrahman Mahmoud, Tianyu Jia, Yu-Shun Hsiao, Vijay Janapa Reddi, and Arijit Raychowdhury  
*Design, Automation and Test in Europe Conference (DATE)*, 2022
- C45. ASP-DAC'22** “Circuit and System Technologies for Energy-Efficient Edge Robotics”  
**Zishen Wan**, Ashwin Lele, and Arijit Raychowdhury  
*Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2022
- C46. DAC'21** “Analyzing and Improving Fault Tolerance of Learning-Based Navigation Systems”  
**Zishen Wan**, Aqeel Anwar, Yu-Shun Hsiao, Tianyu Jia, Vijay Janapa Reddi, and Arijit Raychowdhury  
*ACM/IEEE Design Automation Conference (DAC)*, 2021  
**Best Presentation Award, DAC Young Fellow Program**
- C47. AICAS'21** “iELAS: An ELAS-Based Energy-Efficient Accelerator for Real-Time Stereo Matching on FPGA Platform”  
Tian Gao, **Zishen Wan**, Yuyang Zhang, Bo Yu, Yanjun Zhang, Shaoshan Liu, and Arijit Raychowdhury  
*IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, 2021
- C48. AICAS'21** “An Energy-Efficient Quad-Camera Visual System for Autonomous Machines on FPGA Platform”  
**Zishen Wan**, Yuyang Zhang, Arijit Raychowdhury, Bo Yu, Yanjun Zhang, and Shaoshan Liu  
*IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, 2021
- C49. DAC'20** “Algorithm-Hardware Co-Design of Adaptive Floating-Point Encodings for Resilient Deep Learning Inference”  
Thierry Tambe, En-Yu Yang, **Zishen Wan**, Yuntian Deng, Vijay Janapa Reddi, Alexander Rush, David Brooks, and Gu-Yeon Wei  
*ACM/IEEE Design Automation Conference (DAC)*, 2020  
**DAC Best Paper Award; ACM SIGDA Research Highlights Nominee**

#### Refereed Journal Publications

- J1. JSSC'26** “A 40-nm Programmable Heterogeneous SoC with 5.625/0.85 MB RRAM/SRAM for Accelerating Neuro-Symbolic AI Models”  
Che-Kai Liu\*, **Zishen Wan\***, Young-Seok Noh, Mohamed Ibrahim, Samuel D. Spetalnick, Tushar Krishna, Win-San Khwa, Ashwin Sanjay Lele, Yu-Der Chih, Meng-Fan Chang, and Arijit Raychowdhury  
*IEEE Journal of Solid-State Circuits (JSSC)*, 2026
- J2. TECS'26** “Cross-Layer Design of Vector-Symbolic Computing: Bridging Cognition and Brain-Inspired Hardware Acceleration”  
Shuting Du, Mohamed Ibrahim, **Zishen Wan**, Luqi Zheng, Boheng Zhao, Zhenkun Fan, Che-Kai Liu, Tushar Krishna, Arijit Raychowdhury, and Haitong Li  
*ACM Transactions on Embedded Computing Systems (TECS)*, 2026
- J3. Frontiers'25** “Breaking the Memory Wall: Next-Generation Artificial Intelligence Hardware”  
Kaushik Roy, Adarsh Kosta, Tanvi Sharma, Shubham Negi, Deepika Sharma, Utkarsh Saxena, Sourjya Roy, Anand Raghunathan, **Zishen Wan**, Samuel Spetalnick, Che-Kai Liu, and Arijit Raychowdhury  
*Frontiers in Science, invited paper*, 2025
- J4. IEEE Micro'25** “Sipping Matcha of Security: A Fireside Chat With Mengjia Yan”  
Jianming Tong and **Zishen Wan**  
*IEEE Micro*, 2025
- J5. TCASAI'24** “Towards Efficient Neuro-Symbolic AI: From Workload Characterization to Hardware Architecture”  
**Zishen Wan**, Che-Kai Liu, Hanchen Yang, Ritik Raj, Chaojian Li, Haoran You, Yonggan Fu, Cheng Wan, Sixu Li, Youbin Kim, Ananda Samajdar, Yingyan (Celine) Lin, Mohamed Ibrahim, Jan M. Rabaey, Tushar Krishna, and Arijit Raychowdhury  
*IEEE Transactions on Circuits and Systems for Artificial Intelligence (TCASAI)*, 2024  
**Best Paper Award, DARPA/SRC JUMP 2.0, 2024; top-cited paper in TCASAI**

- J6. JATS'24** “**Benchmarking Test-Time DNN Adaptation at Edge with Compute-In-Memory**”  
Zhenkun Fan\*, **Zishen Wan\***, Che-Kai Liu, Anni Lu, Kshitij Bhardwaj, and Arijit Raychowdhury  
*ACM Journal on Autonomous Transportation Systems (JATS)*, 2024
- J7. CACM'24** “**The Vulnerability-Adaptive Protection Paradigm Toward Reliable Autonomous Machines**”  
**Zishen Wan**, Yiming Gan, Bo Yu, Shaoshan Liu, Arijit Raychowdhury, and Yuhao Zhu  
*Communications of the ACM*, 2024
- J8. TCAD'23** “**Silent Data Corruption in Robot Operating System: A Case for End-to-End System-Level Fault Analysis Using Autonomous UAVs**”  
Yu-Shun Hsiao\*, **Zishen Wan\***, Tianyu Jia, Radhika Ghosal, Abdulrahman Mahmoud, Arijit Raychowdhury, David Brooks, Gu-Yeon Wei, and Vijay Janapa Reddi  
*IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2023
- J9. JSSC'23** “**A Heterogeneous RRAM In-Memory and SRAM Near-Memory SoC for Fused Frame and Event-Based Target Identification and Tracking**”  
Ashwin Lele, Muya Chang, Samuel Spetalnick, Brian Crafton, Shota Konna, **Zishen Wan**, Ashwin Bhat, Win-San Khwa, Yu-Der Chih, Meng-Fan Chang, and Arijit Raychowdhury  
*IEEE Journal of Solid-State Circuits (JSSC)*, 2023
- J10. TMLR'22** “**QuaRL: Quantization for Fast and Environmentally Sustainable Reinforcement Learning**”  
Srivatsan Krishnan, Max Lam, Sharad Chitlangia, **Zishen Wan**, Gabriel Barth-Maron, Aleksandra Faust, and Vijay Janapa Reddi  
*Transactions on Machine Learning Research (TMLR)*, 2022
- J11. CAS-M'21** “**A Survey of FPGA-Based Robotic Computing**”  
**Zishen Wan**, Bo Yu, Thomas Li, Jie Tang, Yuhao Zhu, Yu Wang, Arijit Raychowdhury, and Shaoshan Liu  
*IEEE Circuits and Systems Magazine*, 2021
- J12. CAL'20** “**The Sky Is Not the Limit: A Visual Performance Model for Cyber-Physical Co-Design in Autonomous Machines**”  
Srivatsan Krishnan, **Zishen Wan**, Kshitij Bhardwaj, Paul Whatmough, Aleksandra Faust, Gu-Yeon Wei, David Brooks, and Vijay Janapa Reddi  
*IEEE Computer Architecture Letters*, 2020  
**Best Paper Award, IEEE Computer Architecture Letters**
- J13. JJAP'19** “**Electrically Tunable Temporal Imaging in a Graphene-Based Waveguide**”  
Peng Xie, Yu Wen, **Zishen Wan**, Xinyu Wang, Jiarui Liu, Wenqiang Yang, Xiaofeng Li, and Yishan Wang  
*Japanese Journal of Applied Physics*, 2019

## TEACHING

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### Instructor and Course Development

- Fall 2026 **Instructor, Columbia University**  
COMS E6998, *AI-Native Computing: Hardware for AI, AI for Hardware*
- Fall 2025 **Course Developer and Teaching Staff, Harvard University**  
CS249r, *Architecture 2.0: Agentic AI for Computing Systems Design*

### Guest Lectures

- Spring 2025 **Georgia Institute of Technology**  
ECE8893, *Parallel Programming for FPGAs*: “Software-Hardware Co-Design for Neuro-Symbolic AI”
- Spring 2025 **Georgia Institute of Technology**  
ECE8803, *Hardware-Software Co-Design for ML Systems*: “Opportunities and Challenges for In-/Near-Memory Computing”
- Spring 2022, 2023 **Georgia Institute of Technology**  
ECE8893, *Parallel Programming for FPGAs*: “FPGA-Based Robotic Computing”
- Fall 2023 **Kennesaw State University**  
EE6900, *Neuromorphic Computing*

## RESEARCH MENTORING

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- 2025-present **Yichong Zhang**, B.S. student, Tsinghua University
- 2024-present **Jiayi Qian**, Ph.D. student, Georgia Tech
- 2023-present **Hanchen Yang**, Ph.D. student, Georgia Tech



2025-2026 **Zijun Wang**, M.S. student, Georgia Tech; next: Samsung Semiconductor  
 2025-2026 **Arnav Ramamoorthy**, B.S. student, BITS Pilani; next: M.S. student, University of Pennsylvania  
 2025-2026 **Arnav Shukla**, B.S. student, IIIT Delhi; next: Qualcomm  
 2025-2025 **Jirong Yang**, B.S. student, University of Michigan; next: Ph.D. student, UT Austin  
 2024-2026 **Chenyu Wang**, M.S. student, Princeton University; next: Ph.D. student, Harvard University  
 2024-2026 **Yuhang Du**, B.S. student, Fudan University; next: Ph.D. student, UCLA  
 2023-2025 **Zhenkun Fan**, M.S. student, Georgia Tech; next: Ph.D. student, Georgia Tech  
 2023-2024 **Ritik Raj**, Ph.D. student, Georgia Tech  
 2022-2024 **Che-Kai Liu**, B.S. student, Zhejiang University; next: Ph.D. student, Georgia Tech  
 2022-2023 **Ying-Hao Wei**, M.S. student, Georgia Tech; next: Micron

## PROFESSIONAL SERVICE

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### Conference and Workshop Leadership

2026 Co-organizer, *Architecture 2.0: Agentic AI for Computing Systems Design*, ASPLOS  
 2026 Co-organizer, *Architecture 2.0: Agentic AI for Computing Systems Design*, ISCA  
 2026 Co-organizer, *Architecture 2.0: Agentic AI for Computing Systems Design*, IISWC  
 2026 Co-organizer, *VisArch: Systems and Architectures for Visual Computing, AR/VR, and Embodied Intelligence*, ISCA  
 2024 Special-session organizer, *Domain-Specific Hardware for Autonomous Machines*, ICCAD  
 2024 Special-session organizer, *Neuro-Symbolic Architecture Meets LLMs*, ESWEEK

### Conference Technical Program Committees

2026 HPCA, ASPLOS, MICRO, DAC, MLSys, ASAP, ICCD, ISLPED, and MLCAD  
 2025 MLSys and ICRA  
 2024 DAC  
 2023 DAC, ESWEEK, and IJCAI

### Artifact Evaluation Committees

2026 HPCA and MLSys  
 2025 HPCA  
 2024 ISCA  
 2023 MICRO, ISCA, ASPLOS, and MLSys  
 2022 MICRO, ASPLOS, and IISWC

### Workshop Program Committees

2026 MLBench Workshop at ASPLOS  
 2025 Arch4EAI at ISCA; SCOPE at ICLR; Lock-LLM at NeurIPS  
 2024 CAV Workshop at ASPLOS

**Journal Reviewer:** IEEE JSSC, IEEE TCAD, IEEE TCAS-I, IEEE TCAS-II, IEEE TBioCAS, IEEE JETCAS, IEEE Micro, IEEE Internet of Things Journal, IEEE CAL, IEEE TIM, ACM JATS, and ACM TCPS

### Research Community Leadership and Outreach

2024-present Steering Committee, Computer Architecture Student Association  
 2025 Panelist, ML and Systems Rising Stars Workshop  
 2024 News and Media Team, IEEE ISSCC  
 2022-2024 Co-founder, MLCommons Resilience and Robustness Research Working Group  
 2022 Plenary panelist, IEEE COMPSAC

## INVITED TALKS, SEMINARS, AND SELECTED PRESENTATIONS

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### “AI-Native Computing Systems: From Computing for AI to AI for Computing”

Aug 2026 Seminar Talk, Harvard University (hosts: Prof. Juncheng Yang and Prof. Minlan Yu)  
 Aug 2026 Berkeley Agentic AI Summit, University of California, Berkeley

### “Architecting Physical Intelligence: Cross-Stack Co-Design from Systems to Silicon”

Jan-May 2026 Seminars talks at ASU, BU, Columbia, CU Boulder, CUHK, Harvard, HKUST, ICT, JHU, MBZUAI, NCSU, NEU, NUS, Peking, Purdue, Rice, TAMU, Tsinghua, UIUC, UMD, UPenn, USC, UT

### “Tailored Computing: Cross-Layer System, Architecture, and Silicon Co-Design for Embodied Autonomous Machines”

Nov 2025 Invited Talk, Tsinghua University NICS-EFC Lab (host: Dr. Zhenhua Zhu), online  
 Oct 2025 MICRO Ph.D. Forum, Seoul, Korea  
 Jun 2025 DAC Ph.D. Forum, San Francisco, CA  
 Jun 2025 Invited Talk, ISCA Workshop on Architecture Support for Embodied AI Systems, online

- Mar 2025 DARPA/SRC JUMP 2.0 CoCoSys Annual Summit, Atlanta, GA
- Feb 2025 Invited Talk, UIUC Coordinated Science Laboratory, Champaign, IL
- Jan 2025 Seminar Talk, University of Washington (host: Prof. Ang Li), Seattle, WA
- Dec 2024 Seminar Talk, Institute of Computing Technology, Chinese Academy of Sciences, online

#### **“System Implications and Opportunities for Compositional Neuro-Symbolic-Probabilistic AI”**

- Sep 2025 Seminar Talk, Georgia Institute of Technology (host: Prof. Alexey Tumanov), Atlanta, GA

#### **“Demystifying Neuro-Symbolic AI via Workload Characterization and Software-Hardware Co-Design”**

- Jul 2025 Seminar Talk, Purdue University (hosts: Prof. Anand Raghunathan and Prof. Kaushik Roy), West Lafayette, IN
- Jul 2025 Seminar Talk, University of Notre Dame (host: Prof. Ningyuan Cao), South Bend, IN
- Apr 2025 Invited Talk, Google (host: Dr. Suvinay Subramanian), online
- Mar 2025 DARPA/SRC JUMP 2.0 CoCoSys Annual Summit, Atlanta, GA
- Jan 2025 Guest Lecture and Computer Architecture Research Seminar, Georgia Tech, Atlanta, GA
- Nov 2024 ACM/SIGMICRO Student Research Competition, Austin, TX
- Aug 2024 Invited Seminar, University of Minnesota Twin Cities, Minneapolis, MN
- May 2024 MLSys Young Professional Symposium, Santa Clara, CA
- May 2024 International Workshop on Neuro-Symbolic Systems, UC Berkeley, Berkeley, CA
- Mar 2024 DARPA/SRC JUMP 2.0 CoCoSys Annual Summit, Atlanta, GA
- Sep 2023 Guest Lecture, Kennesaw State University EE6900, Atlanta, GA
- May 2023 CoCoSys Annual Summit and Georgia Tech 3D Systems Packaging Research Center, Atlanta, GA

#### **“System-Architecture-Technology Cross-Layer Design for Embodied Intelligence”**

- Nov 2024 Invited Talk, Harvard Nano-Design Research Group (host: Prof. Gage Hills), Cambridge, MA
- Aug 2024 Invited Talk, Lawrence Livermore National Laboratory (host: Dr. Kshitij Bhardwaj), CA

#### **“Intelligence in Robotic Computing: Agile Design Flows for Efficient Autonomous Systems”**

- Nov 2024 Invited Talks, University of Central Florida and MICRO Workshop on Robotic Hardware Acceleration
- Sep 2024 ESWeek Ph.D. Forum, Raleigh, NC
- May 2024 Cyber-Physical Systems Rising Stars Workshop, University of Virginia, Charlottesville, VA
- May 2024 DARPA/SRC JUMP 2.0 CoCoSys Liaison Meeting, Atlanta, GA
- Feb 2024 Career, Research, and Innovation Development Conference, Atlanta, GA
- Nov 2023 IBM AI Compute Symposium, IBM T. J. Watson Research Center, Yorktown Heights, NY
- Aug 2023 ML and Systems Rising Stars Workshop, Google, Mountain View, CA
- May 2023 Georgia Tech Chips Day, Atlanta, GA
- Mar 2023 Georgia Tech EIC Lab, Atlanta, GA
- Feb 2023 CRNCH Annual Summit, Atlanta, GA
- Nov 2022 ACM Student Research Competition at ICCAD, San Diego, CA

#### **“Efficient Algorithm-Hardware Co-Design for Robotic Mapping and Localization”**

- Mar 2023 Career, Research, and Innovation Development Conference, Atlanta, GA
- Oct 2022 IBM AI Compute Symposium, Yorktown Heights, NY
- Oct 2022 CBRIC Annual Summit, Purdue University, West Lafayette, IN
- Feb 2022 CRNCH Annual Summit, Atlanta, GA

#### **“Enabling Reliable and Safe Autonomous Systems”**

- Mar 2024 DARPA/SRC JUMP 2.0 CoCoSys Annual Summit, Atlanta, GA
- Feb 2024 CRNCH Annual Summit, Atlanta, GA
- May 2023 DARPA/SRC JUMP 2.0 CoCoSys Annual Summit, Atlanta, GA
- Jun 2022 IEEE COMPSAC Plenary Panel, online
- Oct 2021 DARPA/SRC JUMP CBRIC Annual Summit, online
- Aug 2021 DARPA/SRC JUMP CBRIC Industry Talk, online
- Jul 2020 Harvard Architecture, Circuits, and Compilers Lab, online

### **BROADER IMPACT AND SELECTED MEDIA**

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#### **Selected Media Coverage**

- Apr 2026 CoCoSys News: “CoCoSys Spotlight: Best 2025 Q3 Publication”
- Feb 2026 Quantum Zeitgeist: “AI’s ‘Thinking’ Speeded Up 681-Fold with New Reasoning Framework”
- Jan 2026 Georgia Tech: “Conference Spotlights Microarchitecture Research and Innovation”
- Oct 2025 Synced: “Collaborative Acceleration: Multi-Robot Cooperation No Longer ‘Half a Beat Slow’!”
- Jul 2025 Georgia Tech: “Wan Wins First Place at Premier Computing Ph.D. Forum”
- May 2025 Georgia Tech: “CoCoSys Develops Groundbreaking Neuro-Symbolic AI Chip”
- Jan 2025 MIT Technology Review: “New Adaptive Protection to Improve Reliability of Robot Computing Systems”

Dec 2024 Fortune: “Generative AI Can’t Shake Reliability Problem, ‘Neuro-Symbolic AI’ Is the Answer”  
Dec 2024 CoCoSys: “Zishen Wan: Research Scholar Spotlight from DARPA/SRC JUMP 2.0”  
Oct 2024 TechXplore: “Balancing Cost and Reliability in Autonomous Machine Design”  
Oct 2024 Georgia Tech: “ECE Students Take Home Top Honors at TECHCON 2024”  
Sep 2024 ACM News: “Hallucination vs. Creativity, and Reliable Autonomous Machines”  
Jun 2024 TechSpot: “Number Representations in Computer Hardware: Fundamentals Matter”  
Apr 2024 Georgia Tech: “ECE Benchmarking Making Major Advances in Machine Learning”  
Jan 2024 Georgia Tech: “Wan Recognized for Energy-Saving Research on Autonomous Systems”  
Oct 2023 Semiconductor Engineering: “Scalable and Compact Multi-Bit CAM Designs Using FeFETs”  
Sep 2023 The Robot Report: “RobotPerf Benchmarks Compare Robotics Computing Performance”  
Jul 2023 Georgia Tech: “Celebrating ISCA’s 50th: Georgia Tech’s Contributions, Impact, and Reflections”  
Dec 2022 Georgia Tech: “Wan Wins Computing Machinery Student Research Competition”